Project – Low Level Design

on

Finance Support Triage Agent

Course Name: Datagami Skill Based Course

Institution Name: Medicaps University

Sr No	Student Name	Enrolment Number
1	ADITYA DUBEY	EN22CS301050
2	ADITYA PARMAR	EN22CS301058
3	ADITYA VYAS	EN22CS301074
4	ALOK DALKE	EN22CS301103
5	ADITYA PRATAP SINGH PARIHAR	EN22CS301061

Group Name: 01D1

Project Number:

Industry Mentor Name:

University Mentor Name:

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1. Introduction

1.1 Scope of the Document

This Low Level Design (LLD) document provides a detailed technical specification of the Finance Support Triage Agent — a production-grade reactive agentic AI system designed to automate the initial response workflow for finance-related communications. The document covers internal architecture, class design, data models, process flows, information flows, entity relationships, and technology justification.

1.2 Intended Audience

Audience	Purpose
Development Team	Implementation reference for classes, APIs, and data flow
Academic Evaluators	LLD presentation covering all five required presentation criteria
Industry / University Mentor	Review of design decisions and technology justifications
QA Team	Understand component boundaries for test case design

1.3 System Overview

The Finance Support Triage Agent is a reactive agentic AI application that processes incoming finance-support messages (emails, chat messages, support tickets) in real time. The agent autonomously:

1. Classifies the message by urgency (Critical / High / Medium / Low) and intent category (Refund, Dispute, Account Query, Fraud Alert, General Enquiry).
2. Extracts named entities such as transaction IDs, account numbers, dates, and monetary amounts using an LLM-powered NER pipeline.
3. Generates a contextually accurate draft reply using Google Gemini 2.5 Flash.
4. Presents results to a human agent via a Gradio web interface for review and dispatch.

> Design Pattern: Reactive Agent — the system perceives inputs (finance messages), reasons via LLM, and acts by producing structured triage output. It does not maintain persistent memory across sessions.

2. System Design

2.1 Application Design — Class Design

The system is decomposed into seven primary classes following the Single Responsibility Principle. Each class has a well-defined role within the pipeline.

2.1.1 Class Inventory

Class Name	Module	Responsibility
TriageRequest	models.py	Pydantic model — validates and structures the incoming finance query payload
TriageResult	models.py	Pydantic model — holds urgency level, intent, extracted entities, and draft response
ExtractedEntities	models.py	Pydantic model — structured container for all NER output fields
ClassificationEngine	classifier.py	Classifies query urgency (Critical/High/Medium/Low) and intent category via Gemini
EntityExtractor	ner_engine.py	Executes NER via Gemini to extract IDs, dates, amounts, and account numbers from text
ResponseGenerator	response_gen.py	Calls Gemini 2.5 Flash to produce a contextual draft response from classification and entities
TriageOrchestrator	agent.py	Central reactive agent — coordinates all three engines and assembles the final TriageResult
GradioUI	app.py	Gradio-based frontend — accepts input, triggers orchestrator, renders structured triage dashboard

2.1.2 Class Descriptions

TriageRequest (models.py)

The entry-point data contract for the entire system. It is a Pydantic BaseModel that validates every incoming finance message before it enters the agent pipeline. It holds the unique message identifier, the raw text body, the source channel (email, chat, or ticket), the timestamp, and an optional metadata dictionary for additional context.

TriageResult (models.py)

The output contract produced at the end of every agent run. It is a Pydantic BaseModel that captures the full triage outcome: the assigned urgency level (Critical, High, Medium, or Low), the detected intent category, the nested ExtractedEntities object, the LLM-generated draft response, a confidence score, and a processing timestamp.

ExtractedEntities (models.py)

A nested Pydantic model embedded inside TriageResult. It holds all Named Entity Recognition (NER) output as typed lists: transaction IDs, account numbers, dates, monetary amounts, and customer identifiers. Having a dedicated model ensures all NER fields are validated and structured before use.

ClassificationEngine (classifier.py)

Responsible solely for determining the urgency and intent of an incoming message. It accepts the raw text, constructs a structured classification prompt, and calls the Gemini 2.5 Flash API. The JSON response is parsed and returned as a ClassificationOutput object containing the urgency level, intent category, and a confidence score.

EntityExtractor (ner_engine.py)

Performs Named Entity Recognition on the raw message text. It builds a precise NER prompt instructing Gemini to return a JSON object containing all identifiable finance entities. The response is parsed and returned as an ExtractedEntities Pydantic model, ready to be embedded in the final TriageResult.

ResponseGenerator (response_gen.py)

Generates the contextual draft reply to send to the customer. It combines the original message, the classification output, and the extracted entities into a rich context prompt, then calls Gemini 2.5 Flash. The returned text is a professional, intent-aware draft response tailored to the specific finance query.

TriageOrchestrator (agent.py)

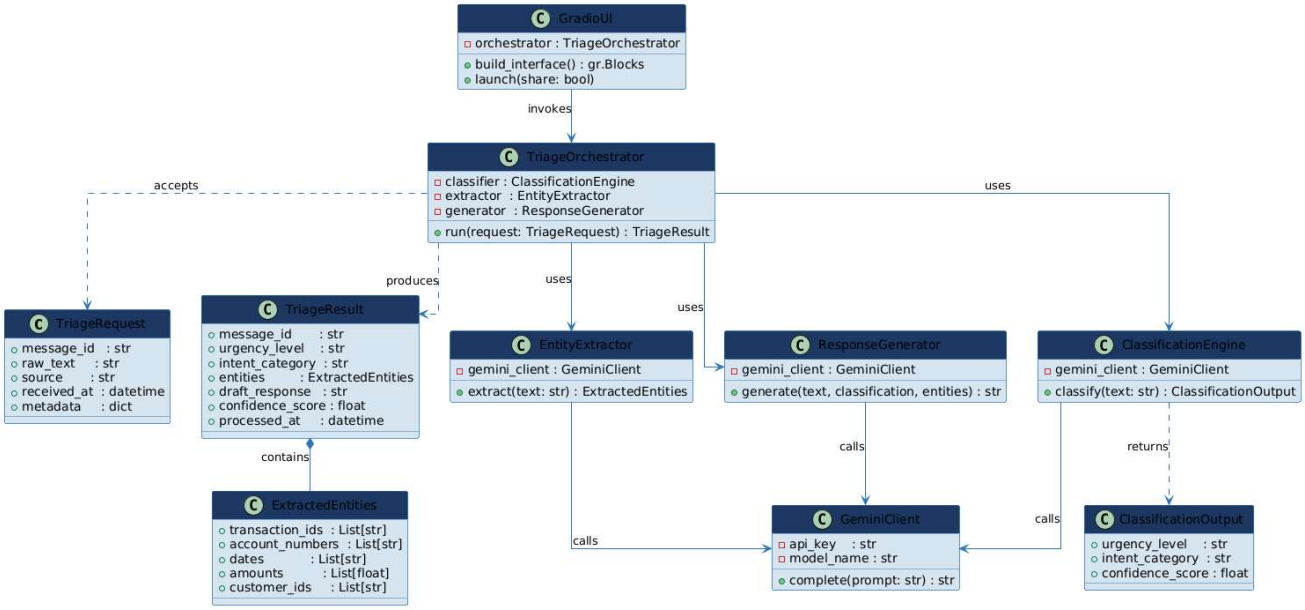
The central reactive agent and the single entry-point for all triage processing. It coordinates the three engines in sequence: first it classifies the message, then it extracts entities, and finally it generates the draft response. All three outputs are assembled into a TriageResult and returned to the UI layer.

GradioUI (app.py)

The frontend layer of the application. It wraps the TriageOrchestrator inside a Gradio Blocks interface, presenting a text input area and a submit button to the human agent. On submission it invokes the orchestrator and renders the structured results in dedicated output panels showing urgency, intent, entities, and the draft response side by side.

2.1.3 Class Diagram

Class Diagram — Finance Support Triage Agent


GradioUI
orchestrator : TriageOrchestrator
build_interface() : gr.Blocks
launch(share: bool)
invokes
TriageOrchestrator
classifier : ClassificationEngine
extractor : EntityExtractor
generator : ResponseGenerator
run(request: TriageRequest) : TriageResult
accepts
uses
produces
uses
uses
TriageRequest
message_id : str
raw_text : str
source : str
received_at : datetime
metadata : dict
TriageResult
message_id : str
urgency_level : str
intent_category : str
entities : ExtractedEntities
draft_response : str
confidence_score : float
processed_at : datetime
EntityExtractor
gemini_client : GeminiClient
extract(text: str) : ExtractedEntities
ResponseGenerator
gemini_client : GeminiClient
generate(text, classification, entities) : str
ClassificationEngine
gemini_client : GeminiClient
classify(text: str) : ClassificationOutput
contains
calls
returns
ExtractedEntities
transaction_ids : List[str]
account_numbers : List[str]
dates : List[str]
amounts : List[float]
customer_ids : List[str]
calls
GeminiClient
api_key : str
model_name : str
complete(prompt: str) : str
calls
ClassificationOutput
urgency_level : str
intent_category : str
confidence_score : float

2.2 Process Flow — User Flow

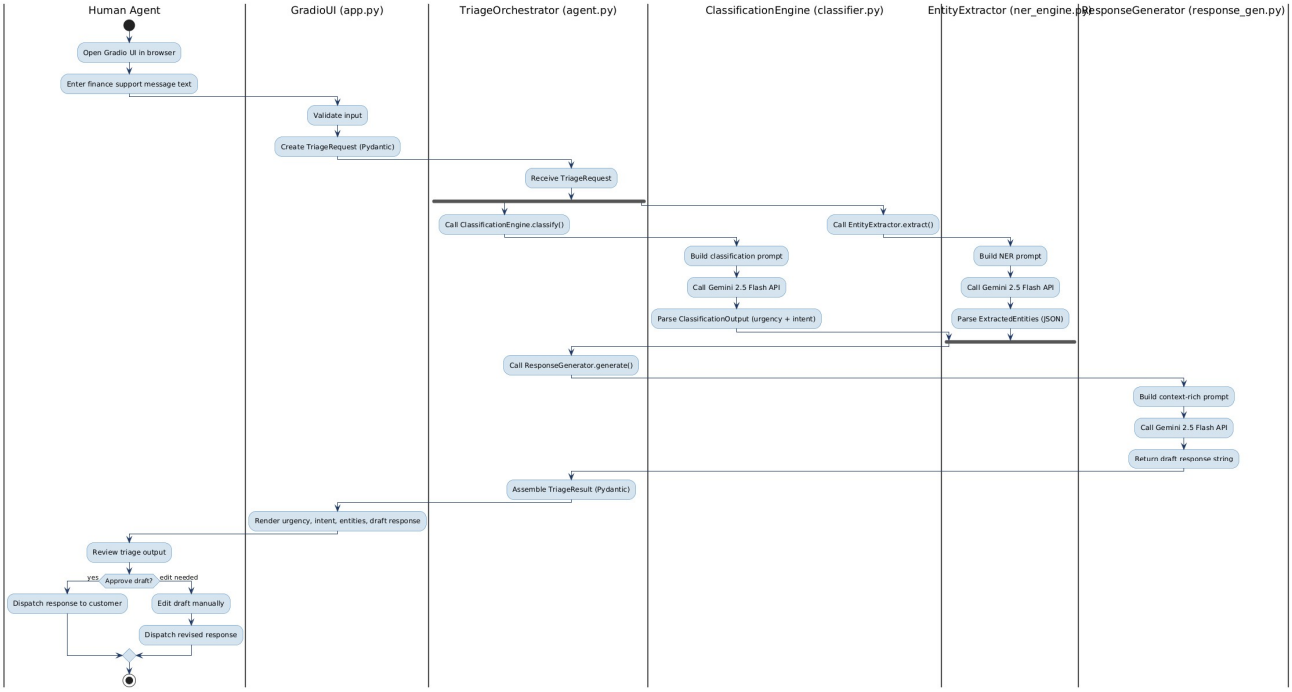
This section describes the complete end-to-end flow from a human agent submitting a finance message to receiving a structured triage result.

2.2.1 User Flow Steps

Step	Actor	Action	Output
1	Human Agent	Opens Gradio UI in browser; enters finance support message text	Text submitted to backend
2	GradioUI	Validates input; creates TriageRequest Pydantic object with metadata	TriageRequest object
3	TriageOrchestrator	Receives TriageRequest; calls ClassificationEngine.classify()	Forwards raw_text to engine
4	ClassificationEngine	Constructs classification prompt; sends to Gemini 2.5 Flash API; parses response	ClassificationOutput (urgency + intent)
5	TriageOrchestrator	Calls EntityExtractor.extract()	Forwards raw_text to NER engine
6	EntityExtractor	Constructs NER prompt; sends to Gemini 2.5 Flash; parses JSON response	ExtractedEntities object
7	TriageOrchestrator	Calls ResponseGenerator.generate() with text + classification + entities	Forwards context to generator
8	ResponseGenerator	Builds context-rich prompt; calls Gemini 2.5 Flash; returns draft response	Draft reply string
9	TriageOrchestrator	Assembles TriageResult Pydantic model from all three outputs	TriageResult object
10	GradioUI	Renders urgency, intent, entities table, and draft response in output panels	Visual triage dashboard
11	Human Agent	Reviews draft; edits if necessary; approves and dispatches to customer	Final response sent to customer

2.2.2 Process Flow Diagram

Process Flow Diagram — Finance Support Triage Agent


Human Agent
GradioUI (app.py)
TriageOrchestrator (agent.py)
ClassificationEngine (classifier.py)
EntityExtractor (ner_engine.py)
ResponseGenerator (response_gen.py)
Open Gradio UI in browser
Enter finance support message text
Validate input
Create TriageRequest (Pydantic)
Receive TriageRequest
Call ClassificationEngine.classify()
Call EntityExtractor.extract()
Build classification prompt
Call Gemini 2.5 Flash API
Parse ClassificationOutput (urgency + intent)
Build NER prompt
Call Gemini 2.5 Flash API
Parse ExtractedEntities (JSON)
Call ResponseGenerator.generate()
Build context-rich prompt
Call Gemini 2.5 Flash API
Return draft response string
Assemble TriageResult (Pydantic)
Render urgency, intent, entities, draft response
Review triage output
Approve draft?
yes
edit needed
Dispatch response to customer
Edit draft manually
Dispatch revised response

2.3 Information Flow

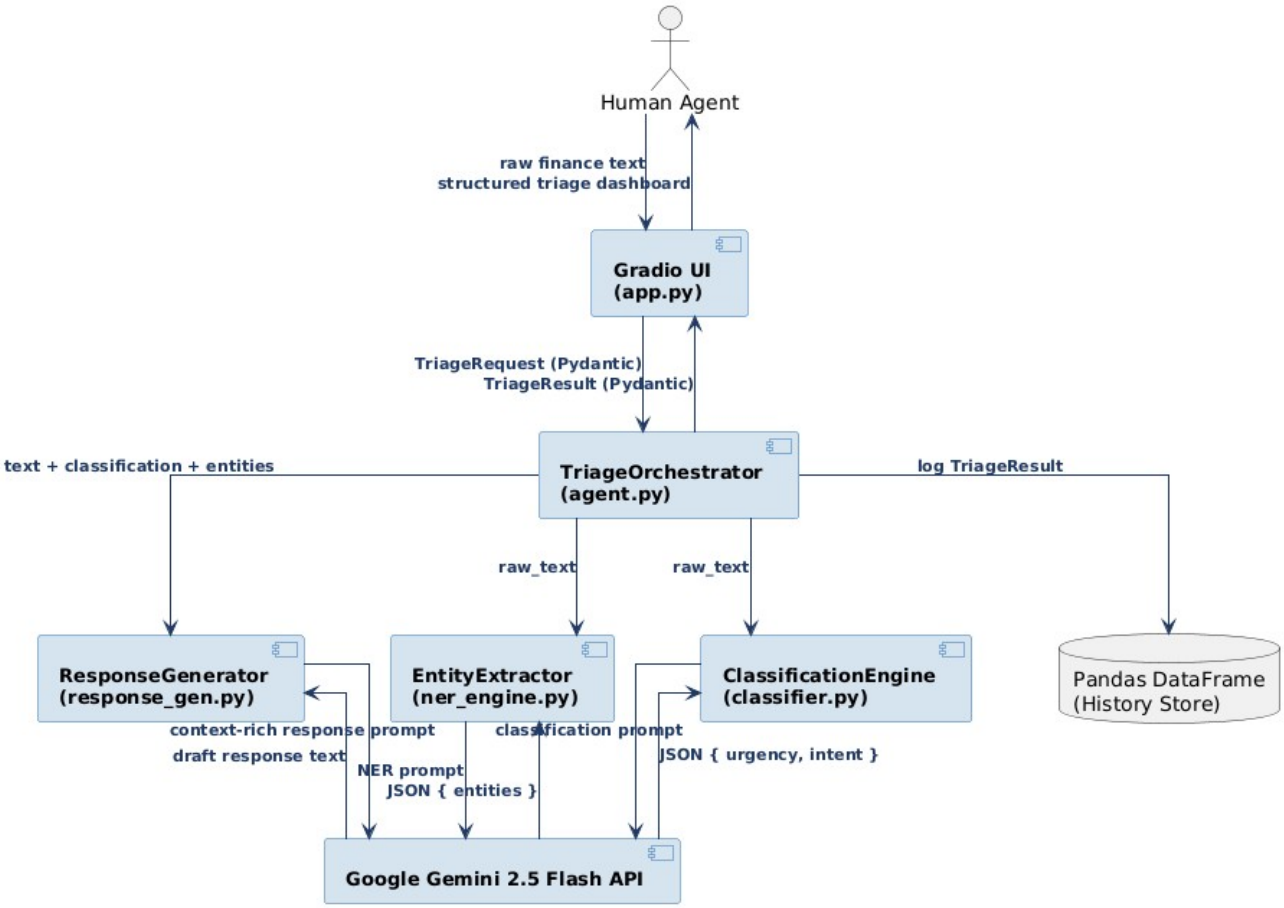
This section details how data transforms as it passes through each component of the system.

2.3.1 Data Transformation Table

Stage	Input	Process	Output	Format
1. Ingestion	Raw finance text (string)	Pydantic validation	TriageRequest object	Pydantic model
2. Classification	TriageRequest.raw_text	LLM prompt → Gemini API	ClassificationOutput	JSON → Pydantic
3. NER Extraction	TriageRequest.raw_text	NER prompt → Gemini API	ExtractedEntities	JSON → Pydantic
4. Response Gen.	text + classification + entities	Context prompt → Gemini API	Draft response string	Plain text
5. Assembly	All three outputs above	TriageResult constructor	TriageResult object	Pydantic model
6. UI Rendering	TriageResult object	Gradio component binding	Human-readable panels	Gradio UI widgets
7. Analytics	Historical TriageResult records	Pandas DataFrame aggregation	Trend report / CSV export	pandas DataFrame

2.3.2 Information Flow Diagram

Information Flow Diagram — Finance Support Triage Agent


Human Agent
raw finance text
structured triage dashboard
Gradio UI
(app.py)
TriageRequest (Pydantic)
TriageResult (Pydantic)
TriageOrchestrator
(agent.py)
text + classification + entities
log TriageResult
raw_text
raw_text
ResponseGenerator
(response_gen.py)
EntityExtractor
(ner_engine.py)
ClassificationEngine
(classifier.py)
Pandas DataFrame
(History Store)
context-rich response prompt
draft response text
NER prompt
JSON { entities }
classification prompt
JSON { urgency, intent }
Google Gemini 2.5 Flash API

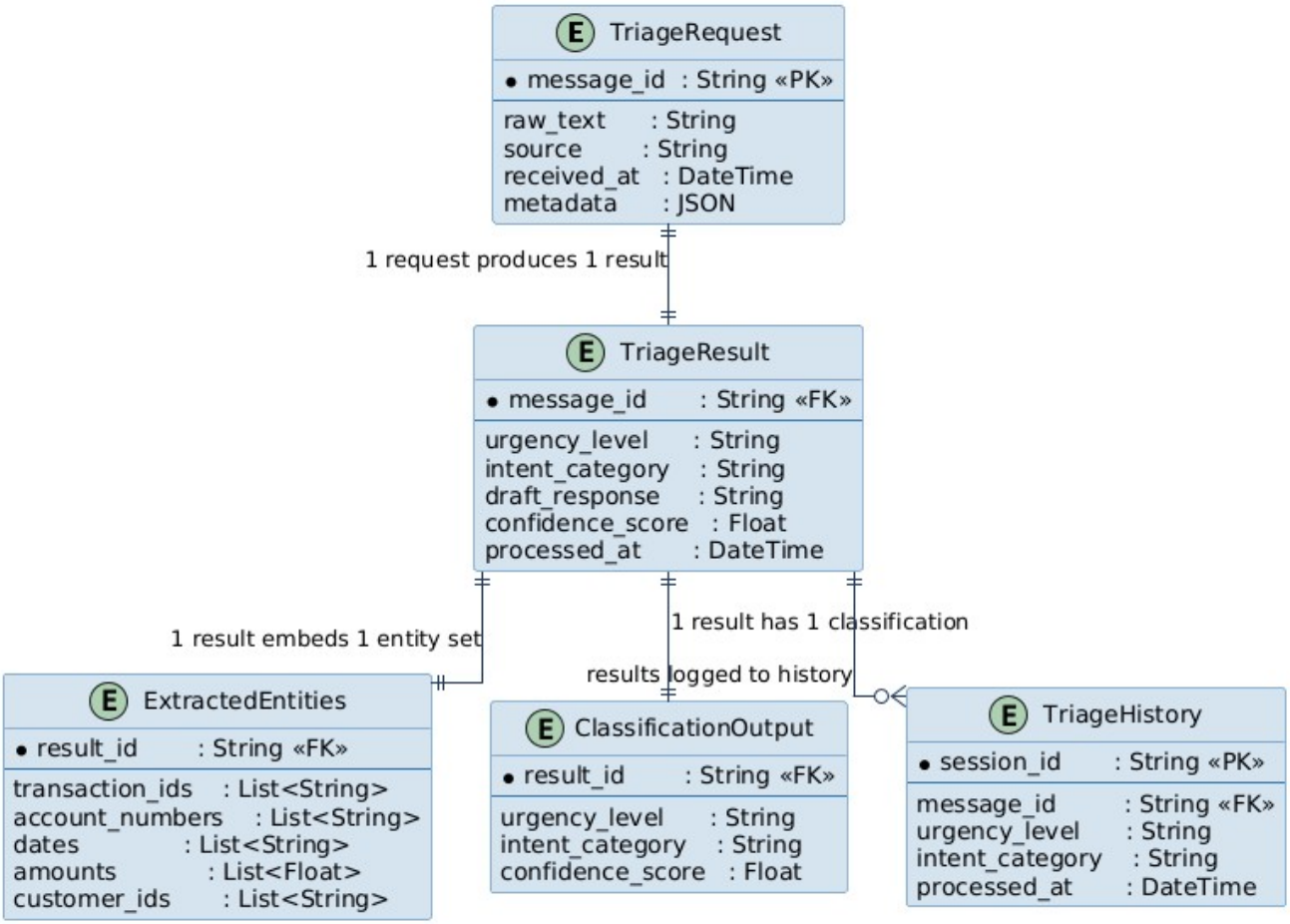
2.4 Components Design — Entities & Relationships

2.4.1 Entity List

Entity	Type	Key Attributes	Role in System
TriageRequest	Pydantic Model	message_id, raw_text, source, received_at, metadata	Input contract; validates all incoming data
TriageResult	Pydantic Model	message_id, urgency_level, intent_category, entities, draft_response, processed_at	Output contract; complete triage output from the agent
ExtractedEntities	Pydantic Model	transaction_ids, account_numbers, dates, amounts, customer_ids	NER output; embedded in TriageResult
ClassificationOutput	Pydantic Model	urgency_level, intent_category, confidence_score	Intermediate result from ClassificationEngine
GeminiClient	Service Client	api_key, model_name	Abstraction layer over Gemini 2.5 Flash HTTP calls
TriageOrchestrator	Agent Controller	classifier, extractor, generator	Orchestrates all engines; single entry-point for UI
TriageHistory	Pandas DataFrame	session_id, message_id, urgency_level, intent_category, processed_at	In-memory analytics store for trend reporting

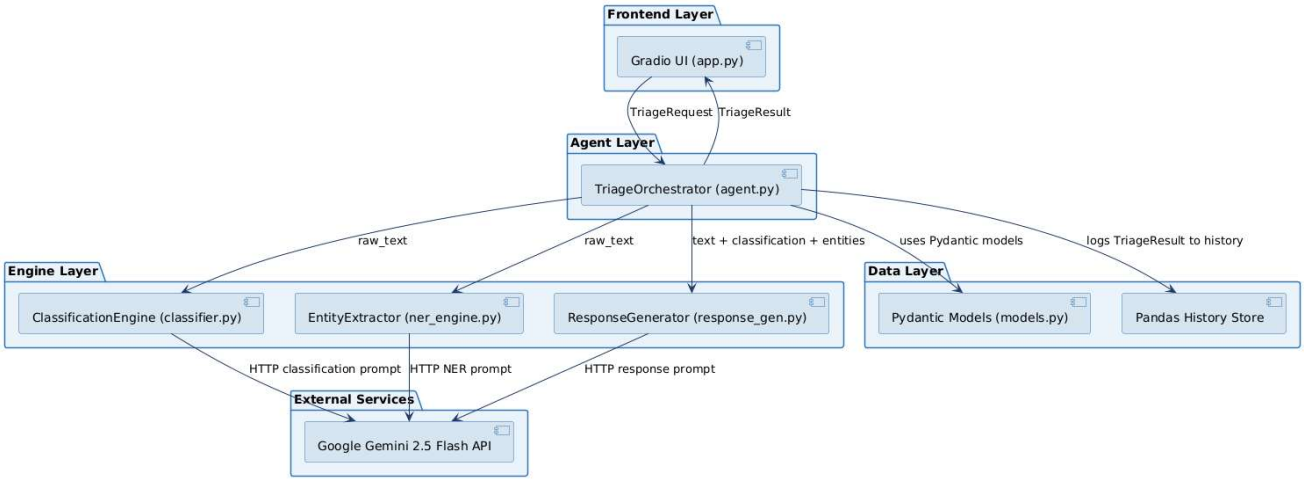
2.4.2 Entity Relationship Diagram

Entity Relationship Diagram — Finance Support Triage Agent



2.4.3 Component Architecture Diagram

Component Architecture Diagram — Finance Support Triage Agent



2.5 Key Design Considerations

2.5.1 Agentic AI Design — Sense–Reason–Act Cycle

The system implements a Reactive Agent architecture. The three-engine design mirrors the Sense–Reason–Act cycle of agentic AI systems:

Agent Phase	Engine	LLM Call Description
SENSE	ClassificationEngine	Perceives message intent and urgency from raw text
REASON	EntityExtractor	Extracts structured facts (entities) from the message
ACT	ResponseGenerator	Generates a contextually accurate action (draft reply)

2.5.2 Error Handling Strategy

- Pydantic validation at ingestion catches malformed inputs before they reach the LLM.
- All Gemini API calls are wrapped in try/except; fallback urgency defaults to 'High' on failure.
- JSON parsing of LLM responses uses safe parsing with default empty models on failure.
- Gradio UI surfaces user-friendly error messages for backend failures.

2.5.3 Scalability Considerations

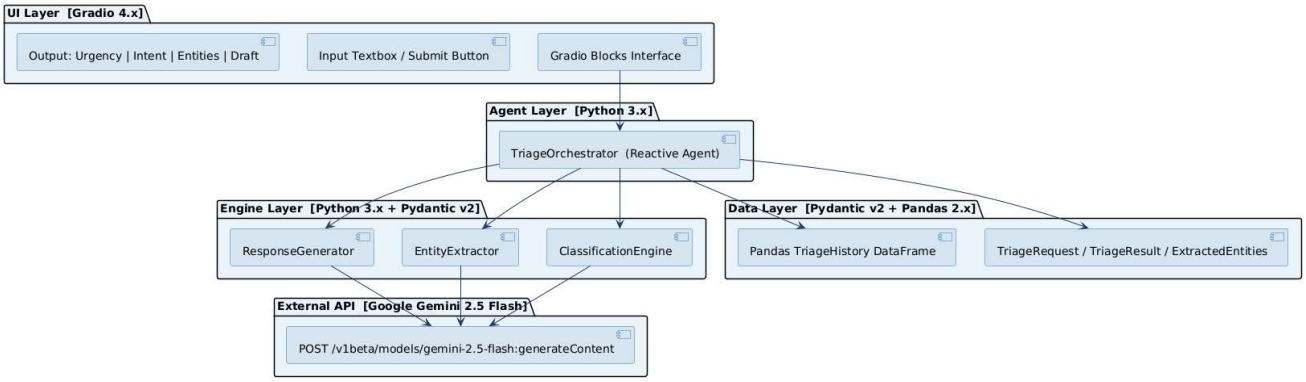
- Stateless design: each TriageRequest is processed independently with no session state.
- GeminiClient is an abstraction layer and can be swapped for any LLM provider (Open/Closed Principle).
- Pandas history store can be replaced with a SQL or NoSQL database for production-scale persistence.
- Gradio async queuing supports multiple concurrent submissions without blocking.

2.6 Technology Stack & Justification

Technology	Version	Role in System	Justification
Python	3.x	Primary language for all backend logic and orchestration	Industry-standard for AI/ML; rich ecosystem for LLM integration, data processing, and web frameworks
Google Gemini 2.5 Flash	API	LLM for classification, NER, and response generation	State-of-the-art multimodal LLM; Flash variant provides low latency ideal for real-time triage; strong structured JSON output capability
Pydantic	v2.x	Data validation and schema enforcement across all models	Enforces strict input/output contracts; eliminates runtime type errors; seamless JSON serialisation and deserialisation between components
Pandas	2.x	Historical triage data processing and analytics reporting	Powerful DataFrame operations for aggregating triage history, computing urgency distributions, and generating trend reports; CSV export support
Gradio	4.x	Interactive web UI for human agent review and dispatch	Zero-boilerplate web interface for Python AI applications; built-in structured output components; rapid prototyping to production deployment

2.6.1 Technology Stack Diagram

Technology Stack — Finance Support Triage Agent



2.7 API Catalogue

2.7.1 Internal Python APIs

Method Signature	Class	Input Type	Output Type	Description
run(request)	TriageOrchestrator	TriageRequest	TriageResult	Main agent entry point; coordinates all engines
classify(text)	ClassificationEngine	str	ClassificationOutput	Returns urgency level and intent category
extract(text)	EntityExtractor	str	ExtractedEntities	Returns all NER entities extracted from text
generate(text,cls,ent)	ResponseGenerator	str, output, entities	str	Returns contextual draft response string
complete(prompt)	GeminiClient	str	str	Abstraction over raw Gemini API HTTP call
build_interface()	GradioUI	—	gr.Blocks	Builds and returns configured Gradio application
launch(share)	GradioUI	bool	None	Starts Gradio development/production server

2.7.2 External API

API	Endpoint	Method	Auth	Called By	Usage
Google Gemini 2.5 Flash	/v1beta/models/gemini-2.5-flash:generateContent	POST	API Key	All 3 Engines	Classification, NER, and response generation inference

3. References

Sr No	Reference	URL / Source
1	Google Gemini API Documentation	https://ai.google.dev/api/generate-content
2	Pydantic v2 Documentation	https://docs.pydantic.dev/latest/
3	Gradio Documentation	https://www.gradio.app/docs
4	Pandas Documentation	https://pandas.pydata.org/docs/
5	PlantUML Language Reference	https://plantuml.com/guide
6	Reactive Agent Architecture	Russell & Norvig, Artificial Intelligence: A Modern Approach, Ch. 2
7	Medicaps University Datagami Course	Internal course material and LLD template